

graythresh Function Test Case Results

Test Case 1: OTSU - Bimodal Image

"Input:" img = uint8([zeros(50,50) 255*ones(50,50)]);

"Output Level: " 0.4980392

"Goodness metric: " 1.

Test Case 2: Mean

"Input:" img = uint8([
 grand(100,1,"uin",20,80);
 grand(100,1,"uin",180,240)
]);

"Output Level: " 0.5172157

Test Case 3: Intermeans

"Input:" img = uint8([
 grand(100,1,"uin",20,70);
 grand(100,1,"uin",180,230)
]);

"Output Level: " 0.4901961

Test Case 4: Intermodes

"Input:" img = uint8([
 10 + grand(100,1,"uin",0,5);
 245 + grand(100,1,"uin",0,5)
]);

"Output Level: " 0.5058824

Test Case 5: MaxEntropy

"Input:" img = uint8([
 30 + grand(100,1,"uin",0,20);
 200 + grand(100,1,"uin",0,20)
]);

"Output Level: " 0.1960784

Test Case 6: MaxLikelihood

"Input:" img = uint8([...
 40 + grand(100,1,"uin",0,20);
 200 + grand(100,1,"uin",0,20)]);

"Output Level: " 0.4901961

Test Case 7: Minimum

"Input:" img = uint8([...
 50 + grand(100,1,"uin",0,10);
 220 + grand(100,1,"uin",0,10)]);

"Output Level: " 0.2235294

Test Case 8: MinError

"Input:" img = uint8([...
 30 + grand(100,1,"uin",0,15);
 210 + grand(100,1,"uin",0,15)]);

"Output Level: " 0.4941176

Test Case 9: Moments

"Input:" img = uint8(grand(128,128,"uin",0,255));

"Output Level: " 0.4980392

Test Case 10: Concavity

"Input:" img = uint8([
 20 + grand(120,1,"uin",0,25);
 120 + grand(120,1,"uin",0,40);
 220 + grand(120,1,"uin",0,15)
]);

"Output Level: " 0.5450980

Test Case 11: Percentile

"Input:" img = uint8([
 grand(150,1,"uin",10,80);
 grand(150,1,"uin",180,240)
]);

"Output Level: " 0.3137255

Test Case 12: Percentile p=0.25

"Input:" img = uint8([
 grand(200,1,"uin",10,60);
 grand(100,1,"uin",180,240)
]);

"Output Level: " 0.1176471

Test Case 13: Percentile p=0.75

"Input:" img = uint8([
 grand(300,1,"uin",10,80);
 grand(100,1,"uin",180,240)
]);

"Output Level: " 0.3137255

Test Case 14: Histogram Input

"Input:" histogram = [10 20 50 100 50 20 10];

"Output Level: " 0.4166667

Test Case 15: Single Bin Histogram

"Input:" histogram = 100;

"Output Level: " 0.

Test Case 16: RGB Image

"Input:" rgb(:, :, 1) = uint8([255*ones(50,25) 50*ones(50,25)]);

rgb(:, :, 2) = uint8([100*ones(50,25) 200*ones(50,25)]);

rgb(:, :, 3) = uint8([0*ones(50,25) 150*ones(50,25)]);

"Output Level: " 0.5

Test Case 17: uint16 Image

"Input:" img = uint16([zeros(50,50) 65535*ones(50,50)]);

"Output Level: " 0.4999924

Test Case 18: Double Image

"Input:" img = rand(100,100);

"Output Level: " 0.5019608

Additional Test Case Cases:

Test Case 1 - Input image:



Output:

otsu (T=0.345)



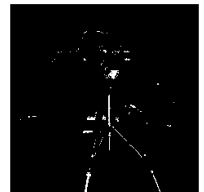
intermeans (T=0.345)



intermodes (T=0.349)



maxentropy (T=0.776)



minimum (T=0.298)



moments (T=0.435)

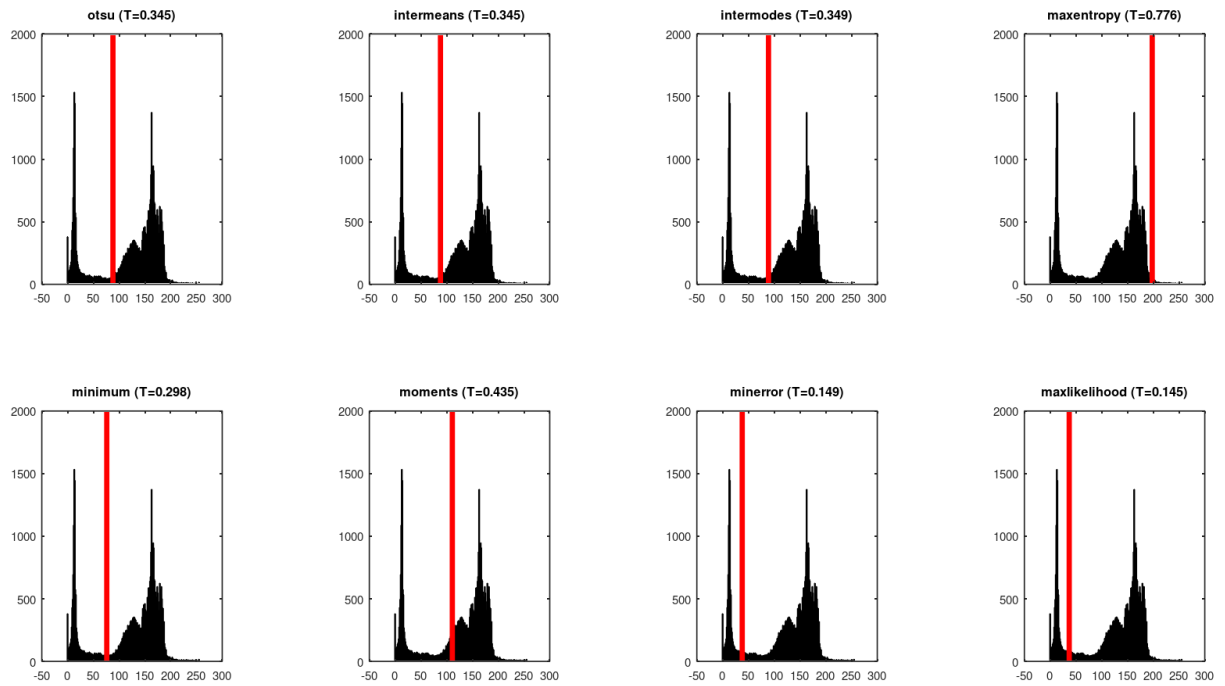


minerror (T=0.149)



maxlikelihood (T=0.145)





Test Case 2 - Input image:



